

Chapter 4:

Imperfections in Solids

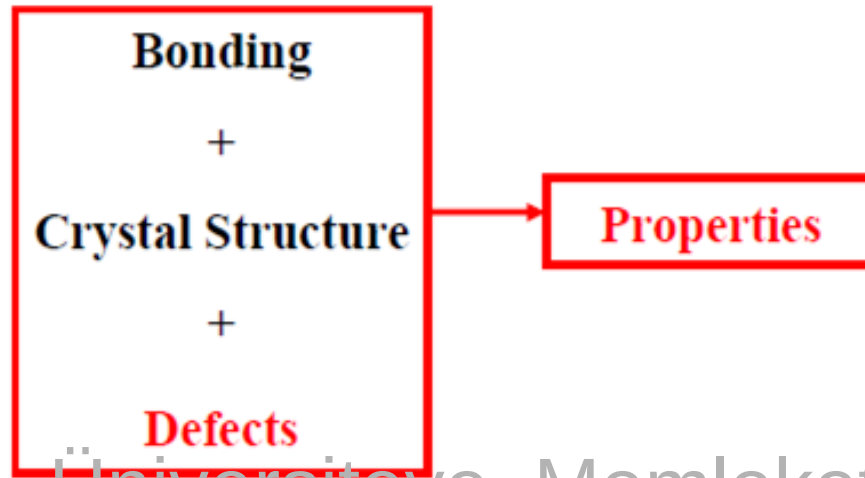
ISSUES TO ADDRESS...

- What are the solidification mechanisms?
- What types of defects arise in solids?
- Can the number and type of defects be varied and controlled?
- How do defects affect material properties?
- Are defects undesirable?

There is **NOT** a such thing as a perfect crystal.

- What are these imperfections?
- Why are they important?

Many of the important properties of materials are due to the presence of imperfections.



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Defects have a profound impact on the various properties of materials:

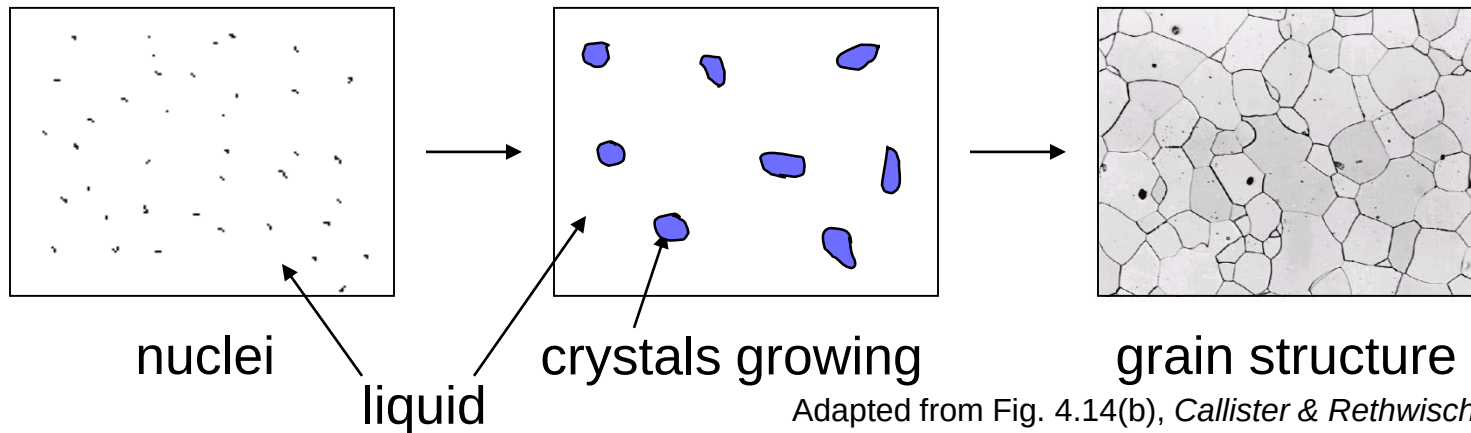
Defects are responsible for color (& price) of a diamond crystal.

Forging a metal tool introduces defects ... and increases strength of the tool.

Imperfections in Solids

- **Solidification** - result of casting of molten material
 - 2 steps
 - Nuclei form
 - Nuclei grow to form crystals – grain structure

- Start with a molten material – all liquid



Adapted from Fig. 4.14(b), Callister & Rethwisch 8e.

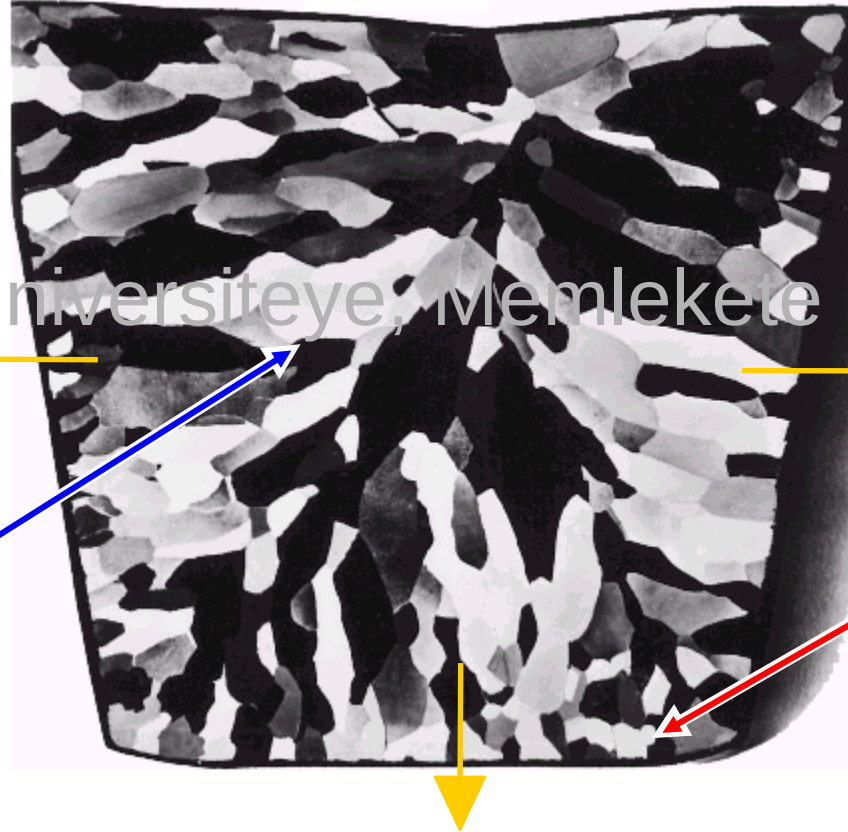
- Crystals grow until they meet each other

Solidification

Grains can be

- equiaxed (roughly same size in all directions)
- columnar (elongated grains)

← ~ 8 cm →



heat
flow

Columnar in
area with less
undercooling

Shell of
equiaxed grains
due to rapid
cooling (greater
 ΔT) near wall

Adapted from Fig. 5.17,
Callister & Rethwisch 3e.

Grain Refiner - added to make smaller, more uniform, equiaxed grains.



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Types of Imperfections

- Vacancy atoms
- Interstitial atoms
- Substitutional atoms

Point defects

- Dislocations

Line defects

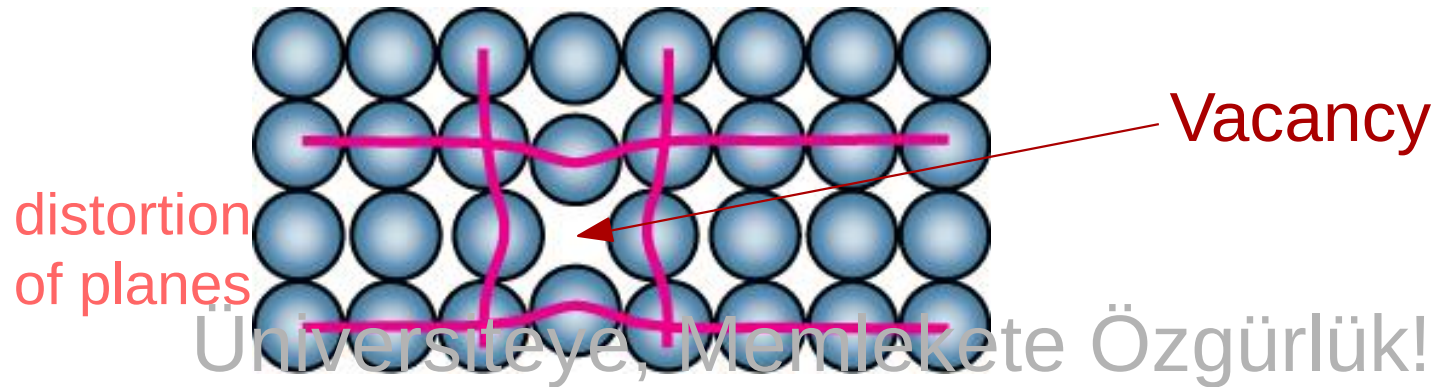
- Grain Boundaries

Area defects

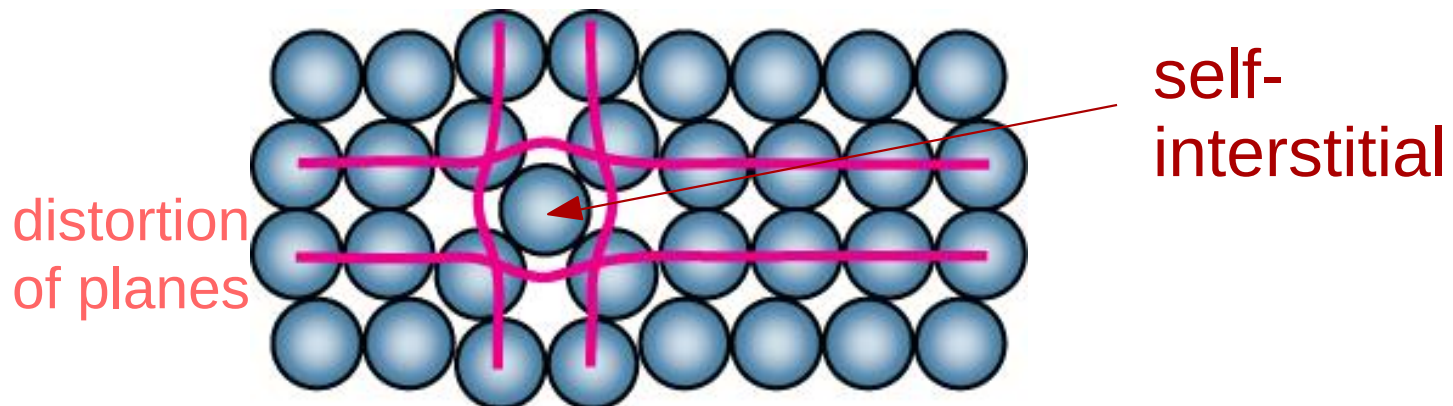
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Point Defects in Metals

- **Vacancies:**
-vacant atomic sites in a structure.



- **Self-Interstitials:**
-"extra" atoms positioned between atomic sites.



Equilibrium Concentration: Point Defects

- Equilibrium concentration varies with temperature!

No. of defects N_v

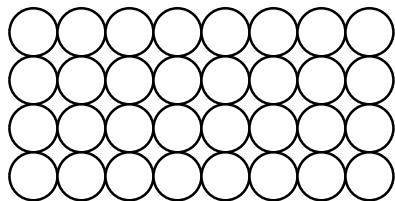
No. of potential defect sites N

Activation energy Q_v

Boltzmann's constant k

Temperature T

$$\frac{N_v}{N} = \exp\left(\frac{-Q_v}{kT}\right)$$



Each lattice site
is a potential
vacancy site

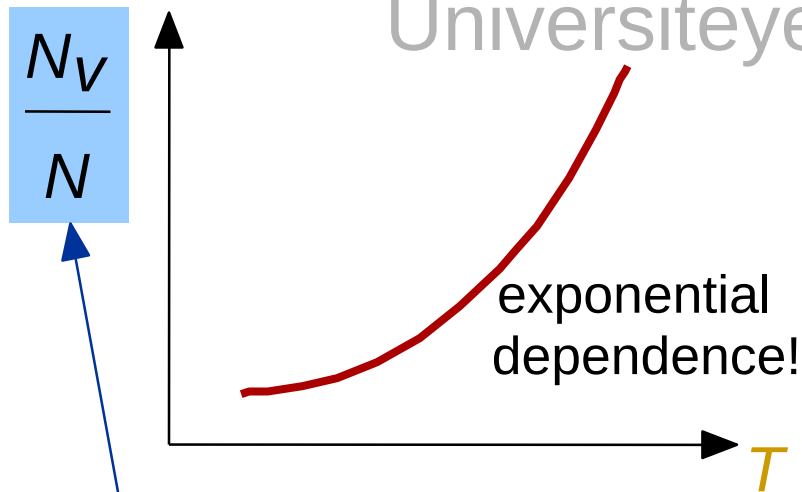
Boltzmann's constant
(1.38×10^{-23} J/atom-K)
(8.62×10^{-5} eV/atom-K)

Measuring Activation Energy

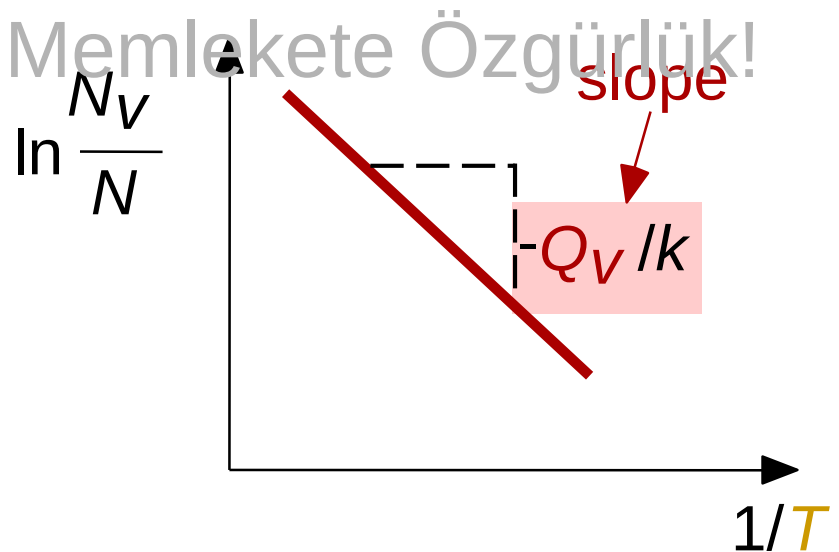
- We can get Q_v from an experiment.
- Measure this...

$$\frac{N_v}{N} = \exp\left(\frac{-Q_v}{kT}\right)$$

- Replot it...



defect concentration



Estimating Vacancy Concentration

- Find the equil. # of vacancies in 1 m³ of Cu at 1000°C.
- Given:

$$\rho = 8.4 \text{ g/cm}^3 \quad A_{\text{Cu}} = 63.5 \text{ g/mol}$$

$$Q_V = 0.9 \text{ eV/atom} \quad N_A = 6.02 \times 10^{23} \text{ atoms/mol}$$

$$\frac{N_V}{N} = \exp\left(\frac{-Q_V}{kT}\right) = 2.7 \times 10^{-4}$$

0.9 eV/atom
1273 K
8.62 x 10⁻⁵ eV/atom-K

For 1 m³, $N = \rho \times \frac{N_A}{A_{\text{Cu}}} \times 1 \text{ m}^3 = 8.0 \times 10^{28} \text{ sites}$

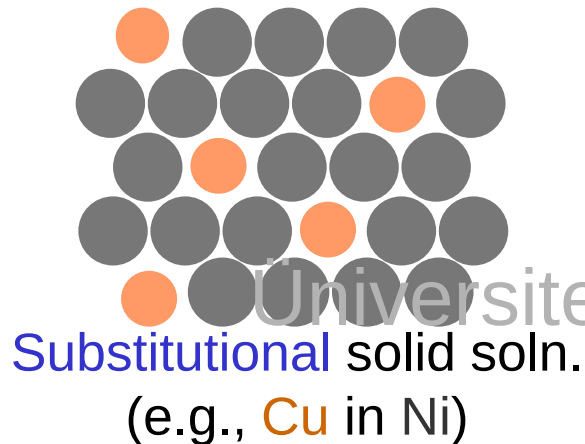
- Answer:

$$N_V = (2.7 \times 10^{-4})(8.0 \times 10^{28}) \text{ sites} = 2.2 \times 10^{25} \text{ vacancies}$$

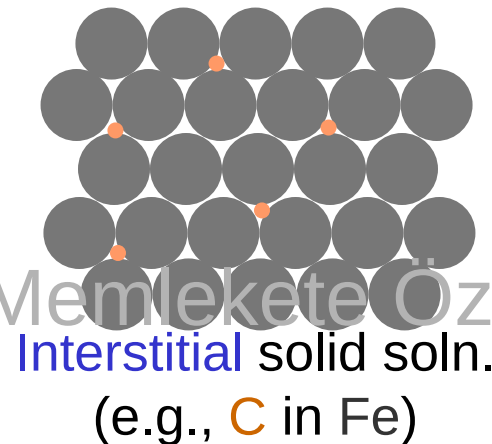
Imperfections in Metals (i)

Two outcomes if impurity (B) added to host (A):

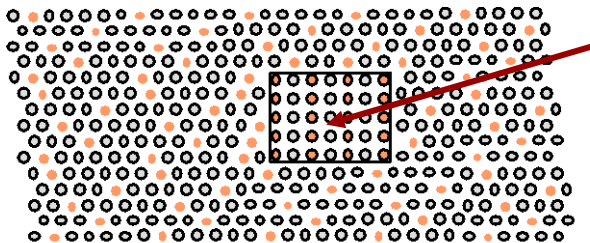
- **Solid solution** of **B** in A (i.e., random dist. of point defects)



OR



- Solid solution of **B** in A plus particles of a new phase (usually for a larger amount of B)



Second phase particle
-- different **composition**
-- often different structure.

Imperfections in Metals (ii)

Conditions for substitutional solid solution (S.S.)

- W. Hume – Rothery rule

- 1. Δr (atomic radius) $< 15\%$
- 2. Proximity in periodic table
 - i.e., similar electronegativities
- 3. Same crystal structure for pure metals
- 4. Valency
 - All else being equal, a metal will have a greater tendency to dissolve a metal of higher valency than one of lower valency

Imperfections in Metals (iii)

Application of Hume–Rothery rules – Solid Solutions

1. Which element does easily solve in Fe ?

2...in Cu?

<i>Element</i>	<i>Atomic Radius (nm)</i>	<i>Crystal Structure</i>	<i>Electro-negativity</i>	<i>Valence</i>
Cu	0.1278	FCC	1.9	+2
C	0.071			
H	0.046			
O	0.060			
Ag	0.1445	FCC	1.9	+1
Al	0.1431	FCC	1.5	+3
Co	0.1253	HCP	1.8	+2
Cr	0.1249	BCC	1.6	+3
Fe	0.1241	BCC	1.8	+2
Ni	0.1246	FCC	1.8	+2
Pd	0.1376	FCC	2.2	+2
Zn	0.1332	HCP	1.6	+2

Table on p. 118, *Callister & Rethwisch 8e.*

Impurities in Solids

- Specification of composition

- weight percent
$$C_1 = \frac{m_1}{m_1 + m_2} \times 100$$

m_1 = mass of component 1.

- atom percent
$$C'_1 = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100$$

n_{m1} = number of moles of component 1

Ex.

What is the composition, in atom percent, of an alloy that consists of 30 wt% Zn and 70 wt% Cu?

In order to compute composition, in atom percent, of a 30 wt% Zn-70 wt% Cu alloy, we employ Equation

$$C'_{\text{Zn}} = \frac{C_{\text{Zn}}A_{\text{Cu}}}{C_{\text{Zn}}A_{\text{Cu}} + C_{\text{Cu}}A_{\text{Zn}}} \times 100$$

$$= \frac{(30)(63.55 \text{ g/mol})}{(30)(63.55 \text{ g/mol}) + (70)(65.41 \text{ g/mol})} \times 100$$

$$= 29.4 \text{ at\%}$$

$$C'_{\text{Cu}} = \frac{C_{\text{Cu}}A_{\text{Zn}}}{C_{\text{Zn}}A_{\text{Cu}} + C_{\text{Cu}}A_{\text{Zn}}} \times 100$$

$$= \frac{(70)(65.41 \text{ g/mol})}{(30)(63.55 \text{ g/mol}) + (70)(65.41 \text{ g/mol})} \times 100$$

$$= 70.6 \text{ at\%}$$

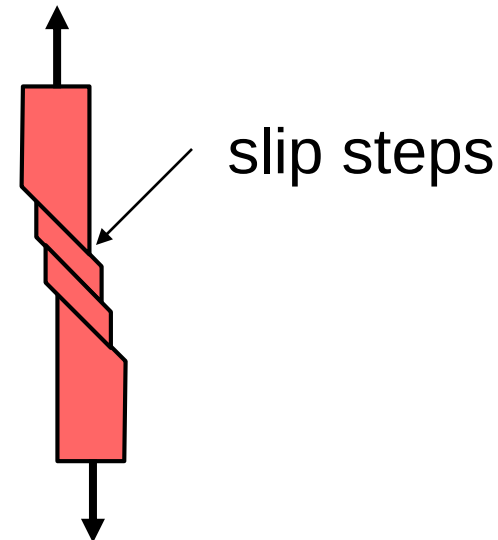
Line Defects

Dislocations:

- are line defects,
- slip between crystal planes result when dislocations move,
- produce permanent (plastic) deformation.

Schematic of Zinc (HCP):

- before deformation
- after tensile elongation



Imperfections in Solids

Linear Defects (Dislocations)

- Are one-dimensional defects around which atoms are misaligned
- **Edge dislocation:**
 - extra half-plane of atoms inserted in a crystal structure
 - **b** perpendicular ($\perp$) to dislocation line
- **Screw dislocation:**
 - spiral planar ramp resulting from shear deformation
 - **b** parallel ($\parallel$) to dislocation line

Burger's vector, $\mathbf{b}$: measure of lattice distortion

Imperfections in Solids

Edge Dislocation

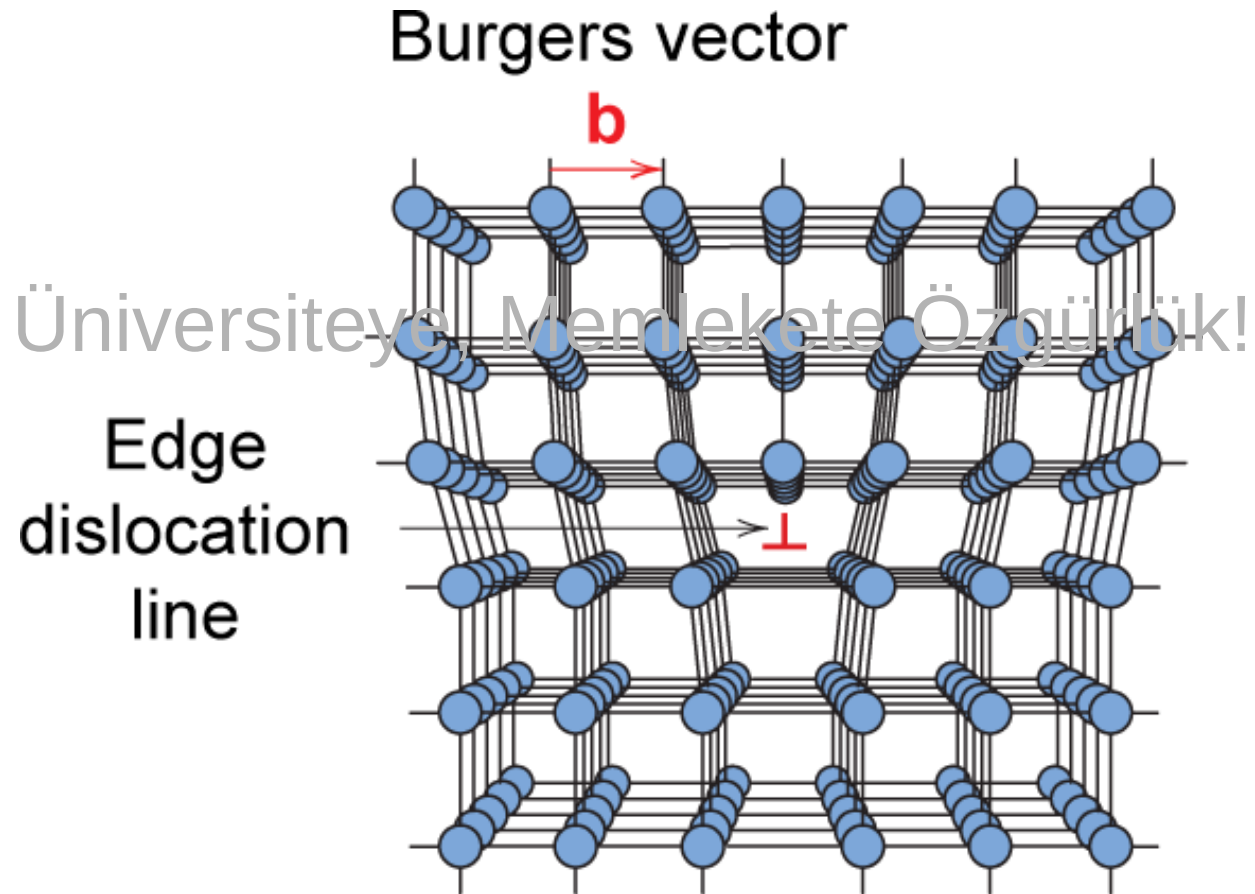
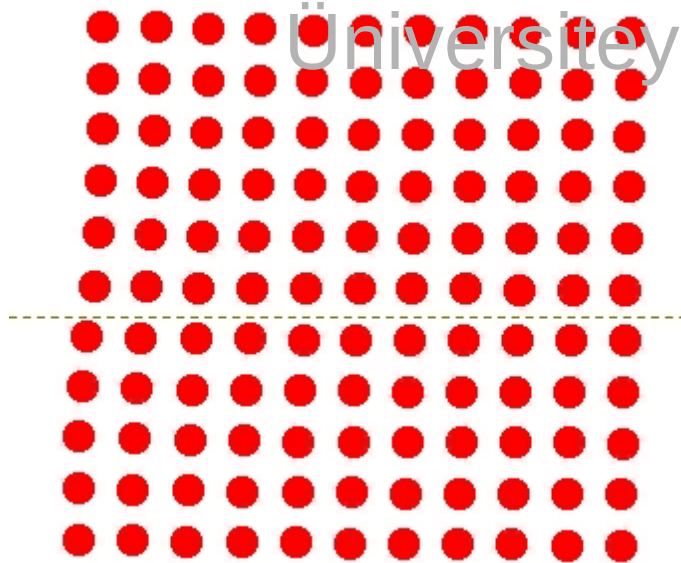


Fig. 4.3, Callister & Rethwisch 8e.

Motion of Edge Dislocation

- Dislocation motion requires the successive bumping of a half plane of atoms (from left to right here).
- Bonds across the slipping planes are broken and remade in succession.



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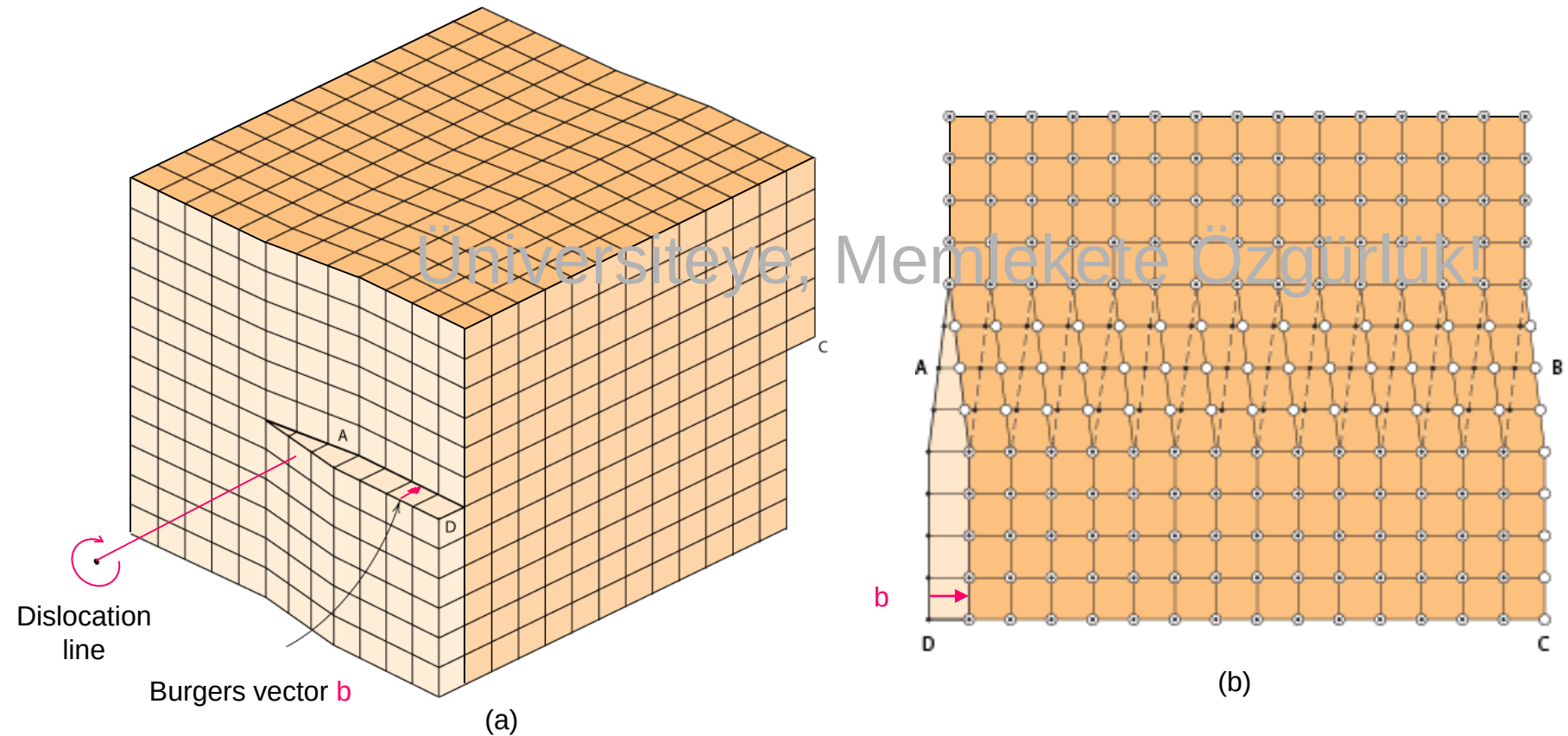
Atomic view of edge dislocation motion from left to right as a crystal is sheared.

Click once on image to start animation

(Courtesy P.M. Anderson)

Imperfections in Solids

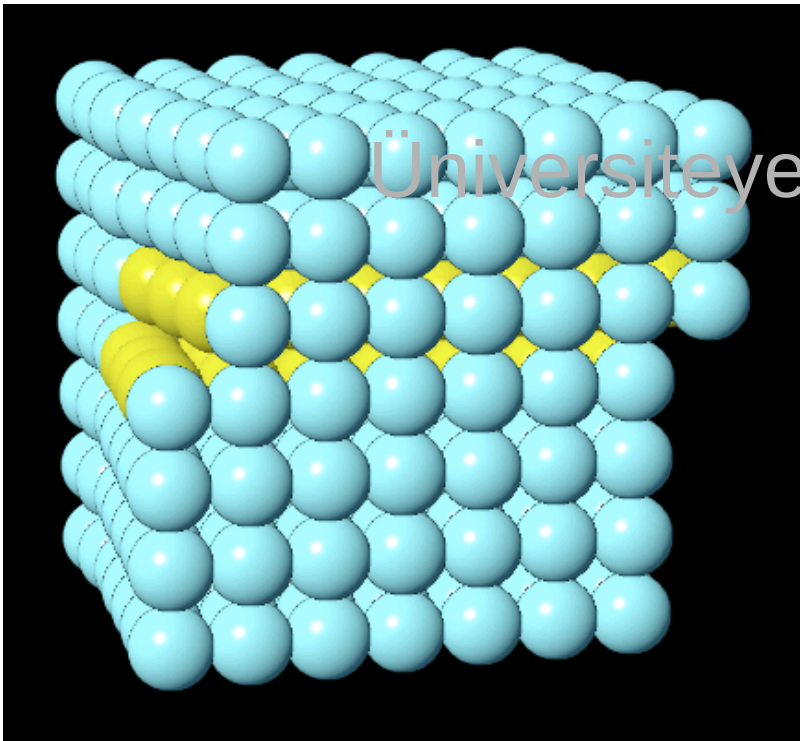
Screw Dislocation



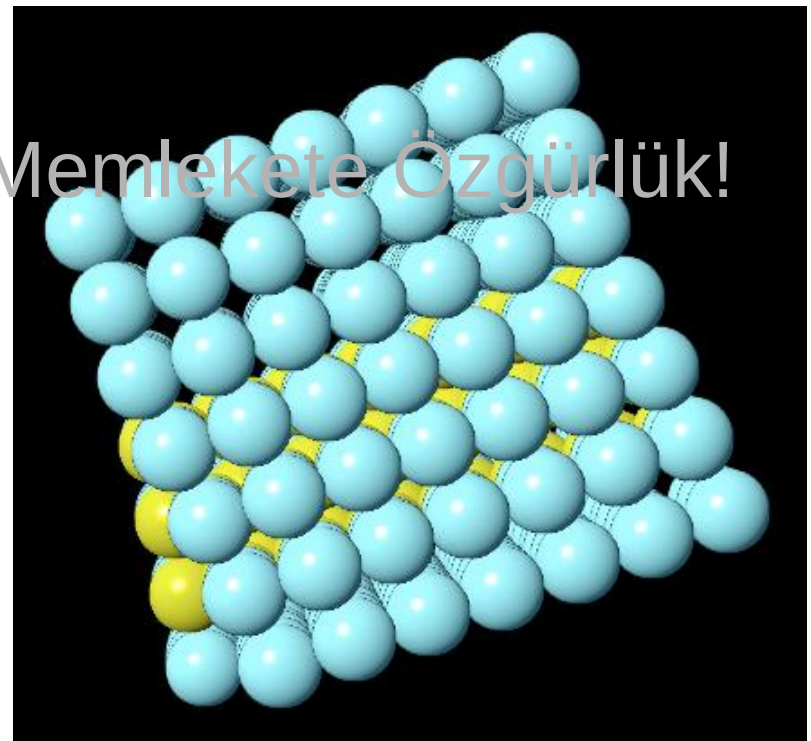
Adapted from Fig. 4.4, *Callister & Rethwisch 8e*.

VMSE: Screw Dislocation

- In VMSE:
 - a region of crystal containing a dislocation can be rotated in 3D
 - dislocation motion may be animated



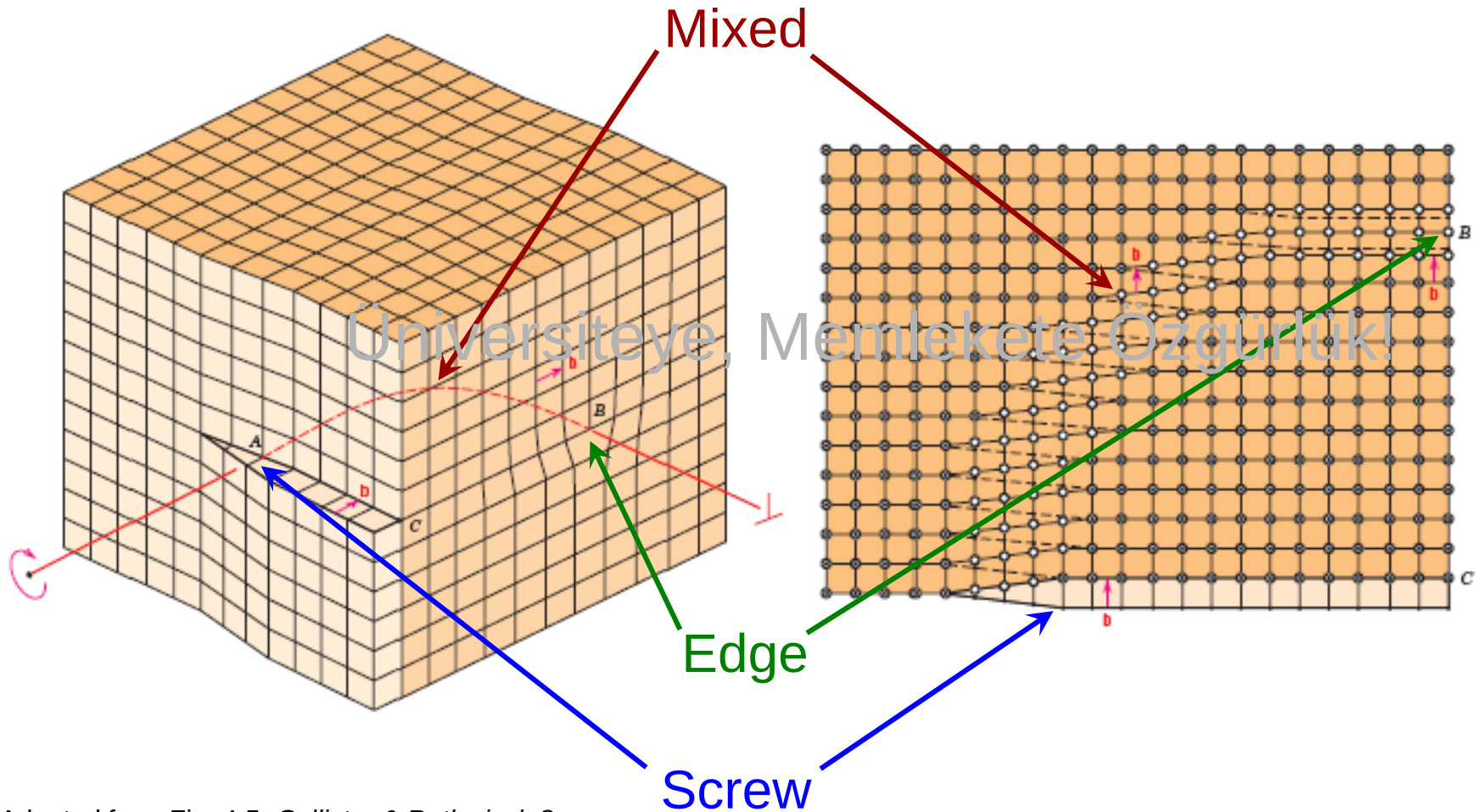
Front View



Top View

VMSE Screen Shots

Edge, Screw, and Mixed Dislocations



Adapted from Fig. 4.5, Callister & Rethwisch 8e.

Imperfections in Solids

Dislocations are visible in electron micrographs

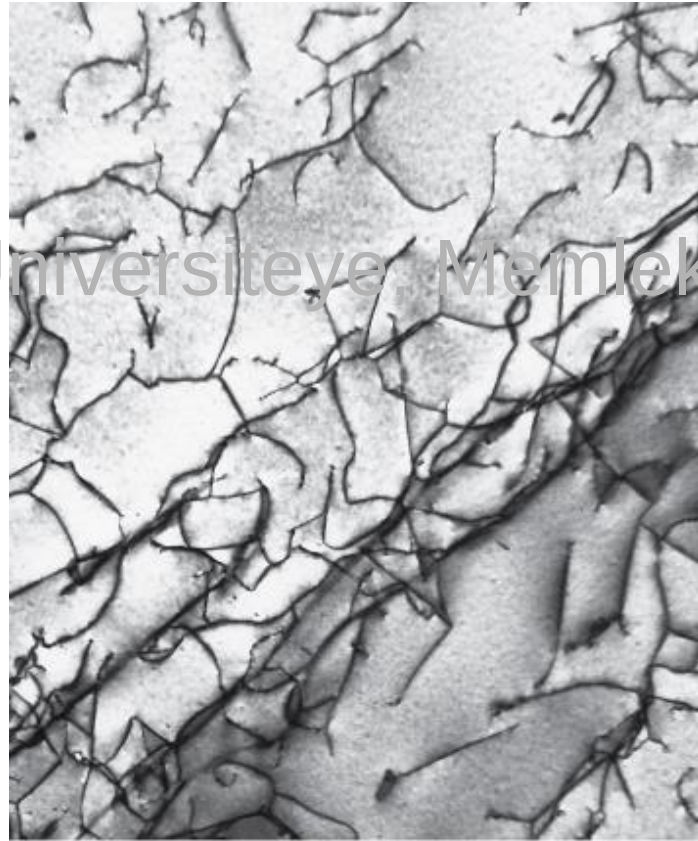


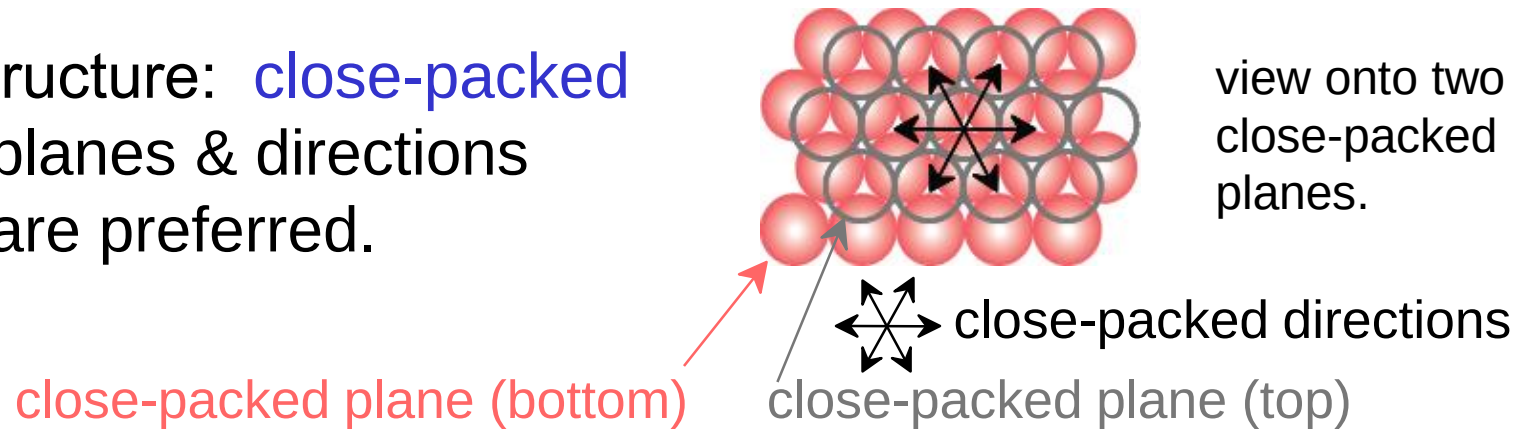
Fig. 4.6, Callister & Rethwisch 8e.

0.2 μm

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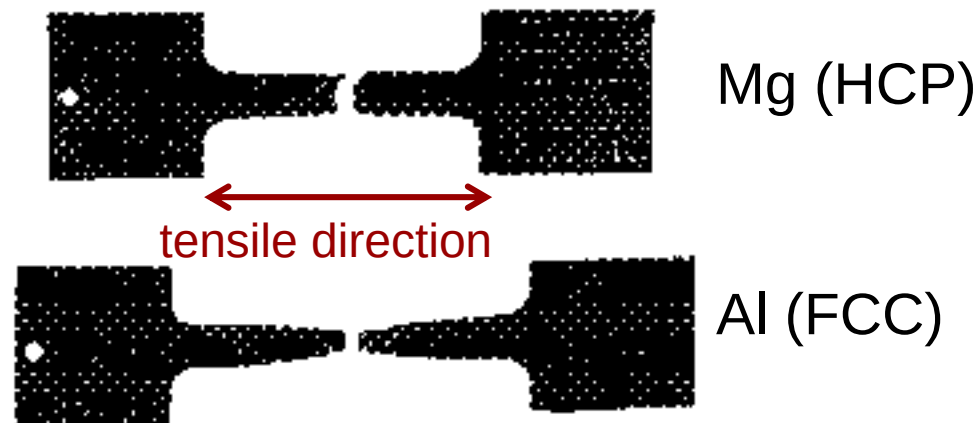
Dislocations & Crystal Structures

- Structure: **close-packed** planes & directions are preferred.



- Comparison among crystal structures:
FCC: many close-packed planes/directions;
HCP: only one plane, 3 directions;
BCC: none

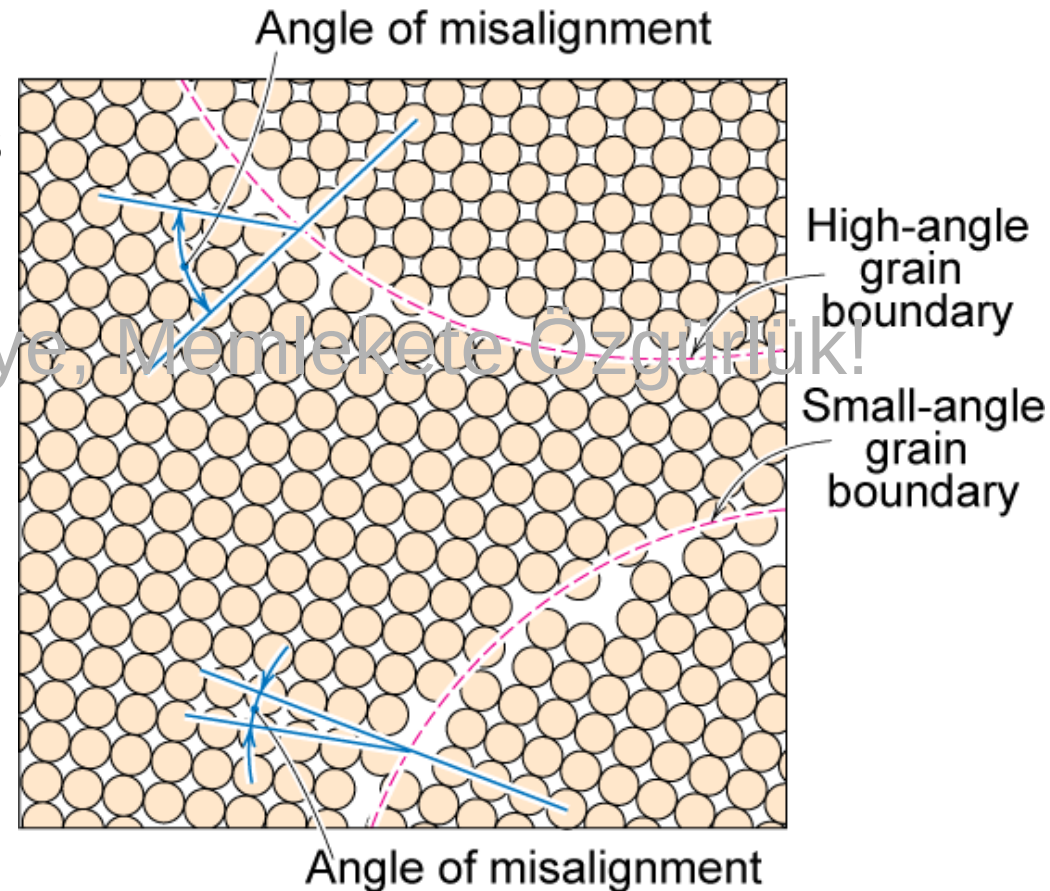
- Specimens that were tensile tested.



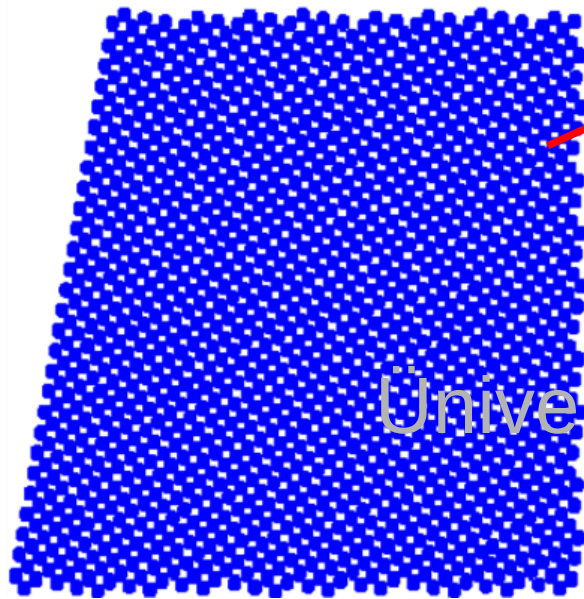
Area defects in solids

Grain Boundaries

- regions between crystals
- transition from lattice of one region to that of the other
- slightly disordered
- low density in grain boundaries
 - high mobility
 - high diffusivity
 - high chemical reactivity



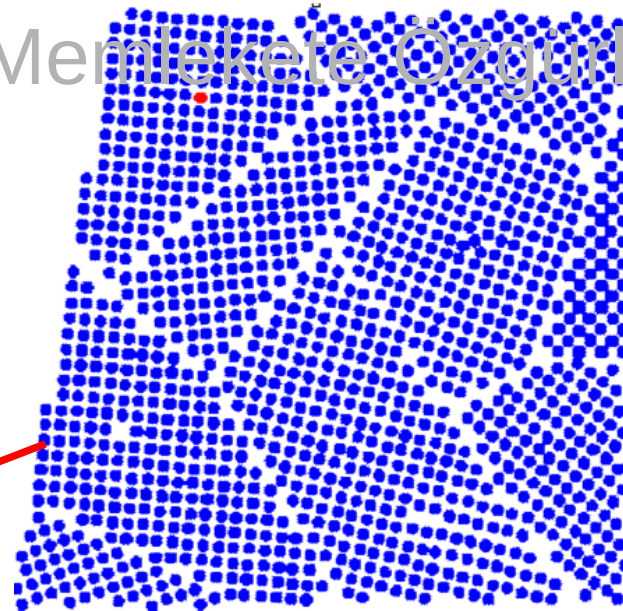
Adapted from Fig. 4.7,
Callister & Rethwisch 8e.



2D representation of a perfect
single crystal atomic arrangement

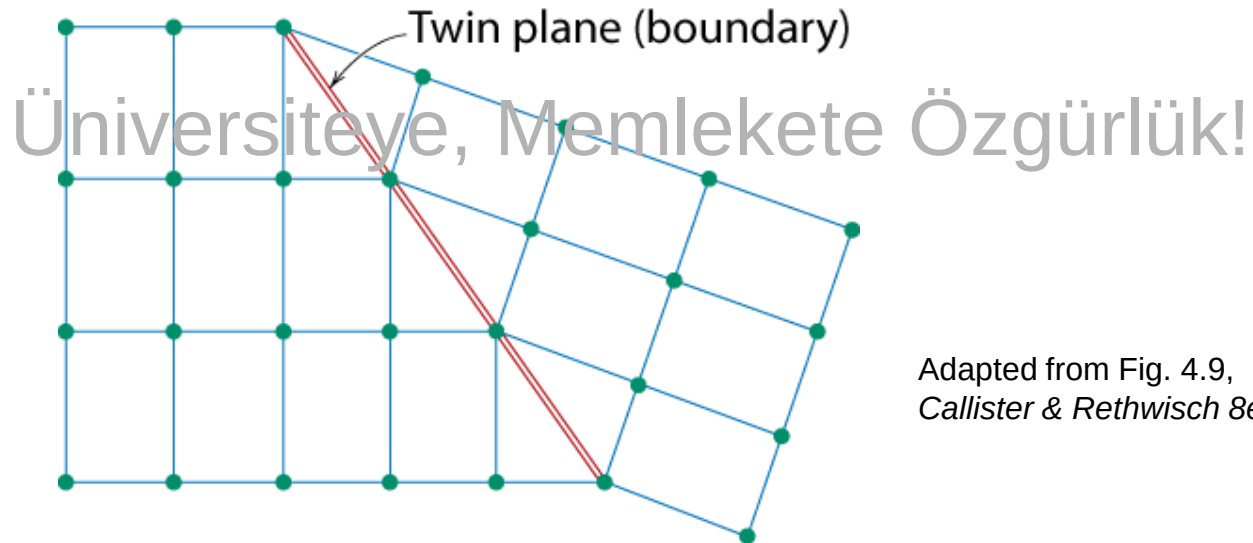
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Structures of real material have
types of defects



Planar Defects in Solids

- One case is a **twin boundary (plane)**
 - Essentially a reflection of atom positions across the **twin plane**.



Adapted from Fig. 4.9,
Callister & Rethwisch 8e.

- **Stacking faults**
 - For FCC metals an error in ABCABC packing sequence
 - Ex: ABCABABC

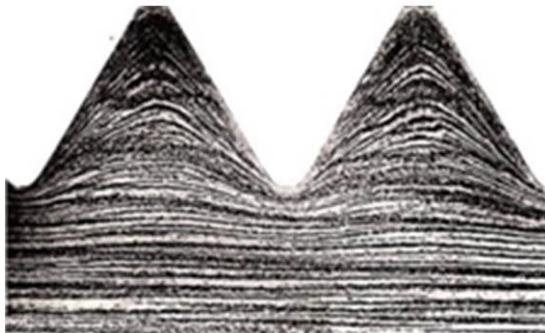
Microscopic Examination

- Crystallites (grains) and grain boundaries.
Vary considerably in size. Can be quite large.
 - ex: Large single crystal of quartz or diamond or Si
 - ex: Aluminum light post or garbage can - see the individual grains
- Crystallites (grains) can be quite small (mm or less) – necessary to observe with a microscope.

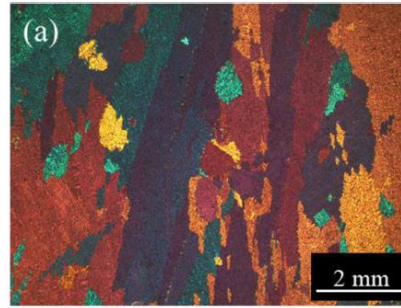
Metallographic sample preparation



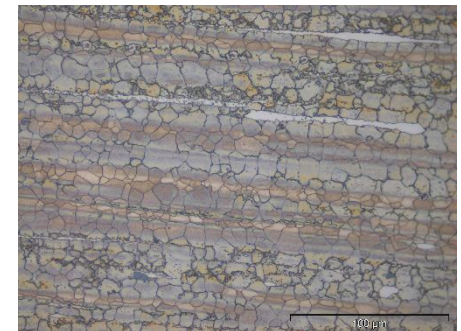
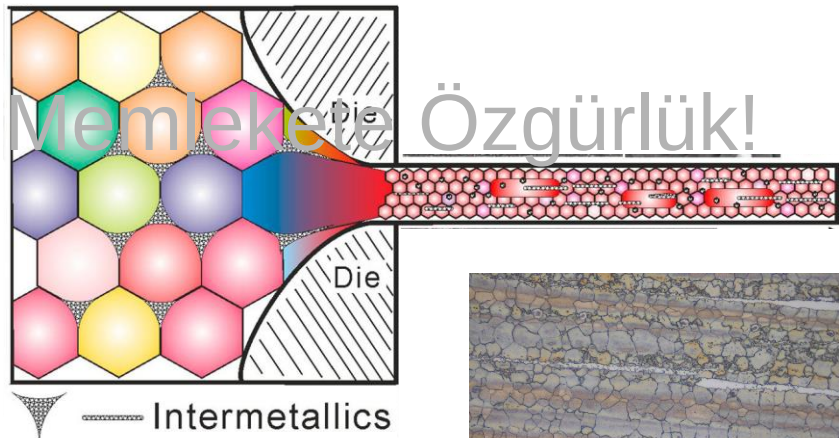
http://www.polycrew.com/20170116_prop.php



<https://www.fastenerdata.co.uk/screw>



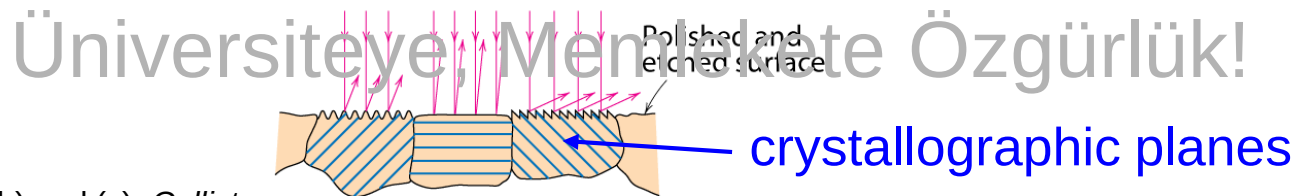
$d = 650 \mu\text{m}$



$d = ? \mu\text{m}$

Optical Microscopy

- Useful up to 2000X magnification.
- Polishing removes surface features (e.g., scratches)
- Etching changes reflectance, depending on crystal orientation.



Adapted from Fig. 4.13(b) and (c), *Callister & Rethwisch 8e*. (Fig. 4.13(c) is courtesy of J.E. Burke, General Electric Co.)



Micrograph of
brass (a Cu-Zn alloy)

← 0.75mm →

Optical Microscopy

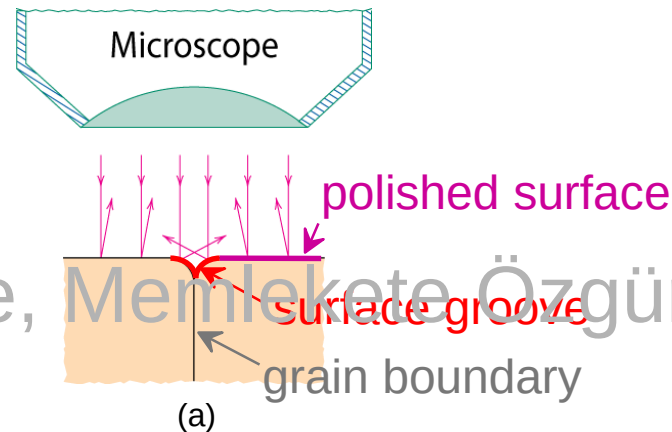
Grain boundaries...

- are imperfections,
- are more susceptible to etching,
- may be revealed as dark lines,
- change in crystal orientation across boundary.

ASTM grain size number

$$N = 2^{n-1}$$

number of grains/in²
at 100x
magnification



(b)

Adapted from Fig. 4.14(a) and (b), *Callister & Rethwisch 8e*.
(Fig. 4.14(b) is courtesy of L.C. Smith and C. Brady, the National Bureau of Standards, Washington, DC [now the National Institute of Standards and Technology, Gaithersburg, MD].)

Optical Microscopy

- Polarized light
 - metallographic scopes often use polarized light to increase contrast
 - Also used for transparent samples such as polymers

Microscopy

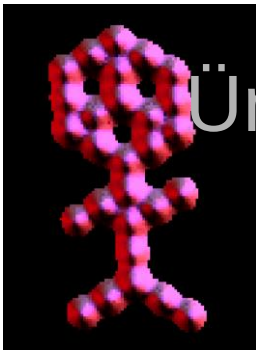
Optical resolution ca. 10^{-7} m = 0.1 μ m = 100 nm

For higher resolution need higher frequency

- X-Rays? Difficult to focus.
- Electrons
 - wavelengths ca. 3 pm (0.003 nm)
 - (Magnification - 1,000,000X)
 - Atomic resolution possible
 - Electron beam focused by magnetic lenses.

Electron Microscopy

- Scanning Electron Microscopy (SEM)
- STM (Scanning Tunneling Microscopy)
Atoms can be arranged and imaged!



Carbon monoxide molecules arranged on a platinum (111) surface.



Iron atoms arranged on a copper (111) surface. These Kanji characters represent the word “atom”.

Photos produced from the work of D.F. Ruiz, Zeppenfeld, and D.M. Eigler. Reprinted with permission from International Business Machines Corporation, copyright 1995.

Summary

- Point, Line, and Area defects exist in solids.
- The number and type of defects can be varied and controlled (e.g., T controls vacancy conc.)
- Defects affect material properties (e.g., grain boundaries control crystal slip).
- Defects may be desirable or undesirable (e.g., dislocations may be good or bad, depending on whether plastic deformation is desirable or not.)